



Background

In a survey by the Pew Research Center, it was found that only a minority of Americans say that they fully read all the privacy policies they encounter when browsing the internet. This leads to a widespread lack of understanding between users and businesses which is only amplified by the fact that most privacy policies exist for the primary purpose of establishing legal grounds for data collection. These documents also aren't actually required by federal level laws in the United States. I created a Google Chrome extension that makes privacy policies easier to read.



Fig 2: Extension Popup

Novelty & Impact

At the time that this project was conceived, a usable privacy policy seeker and summarizer that works on the page and in real time wasn't available. This extension works on demand, combining the common AI Chrome extension and a focus on privacy policy summarization.

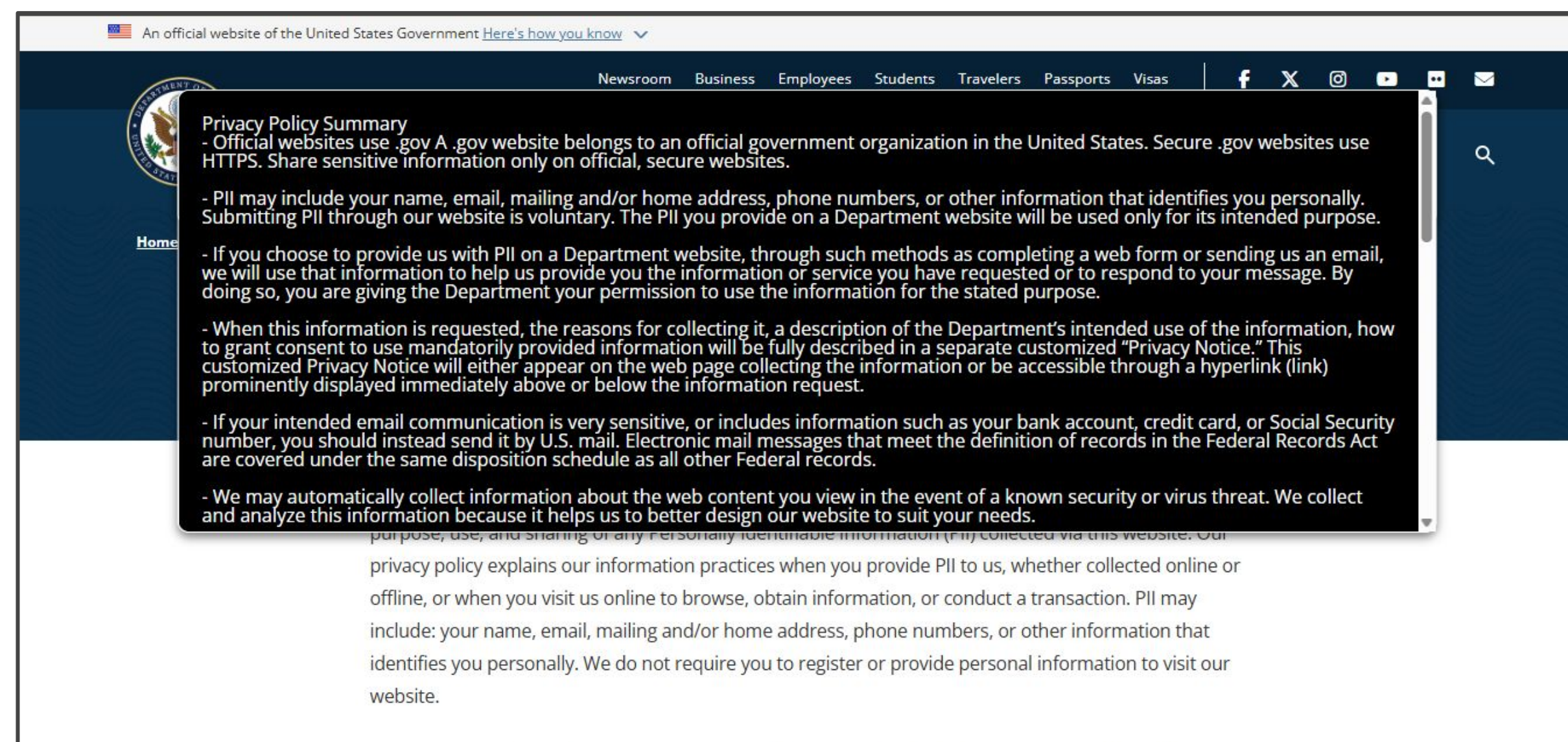


Fig 3: Summary

Methods

The summarization for the extension runs in Runpod where a pretrained BART model from Facebook is being used (Fig. 1). The model combines a bidirectional encoder, like Google's BERT with a left-to-right decoder, similar to OpenAI's GPT models. The front end of the application resembles a standard webpage and uses JS, HTML, and CSS to show an interface (Fig. 2). It extracts the body text from a webpage to send to the RunPod endpoint which queues requests and starts up "workers" to run the Python code that I provided in a container. This is an efficient way to run methods that aren't used constantly.

Results

The extension allows for the extraction of text from webpages of a variety of styles and structures with the click of a button. It creates easy-to-read summaries quickly and efficiently while not disrupting the user's normal browsing (Fig. 3). It also includes limited search capabilities.

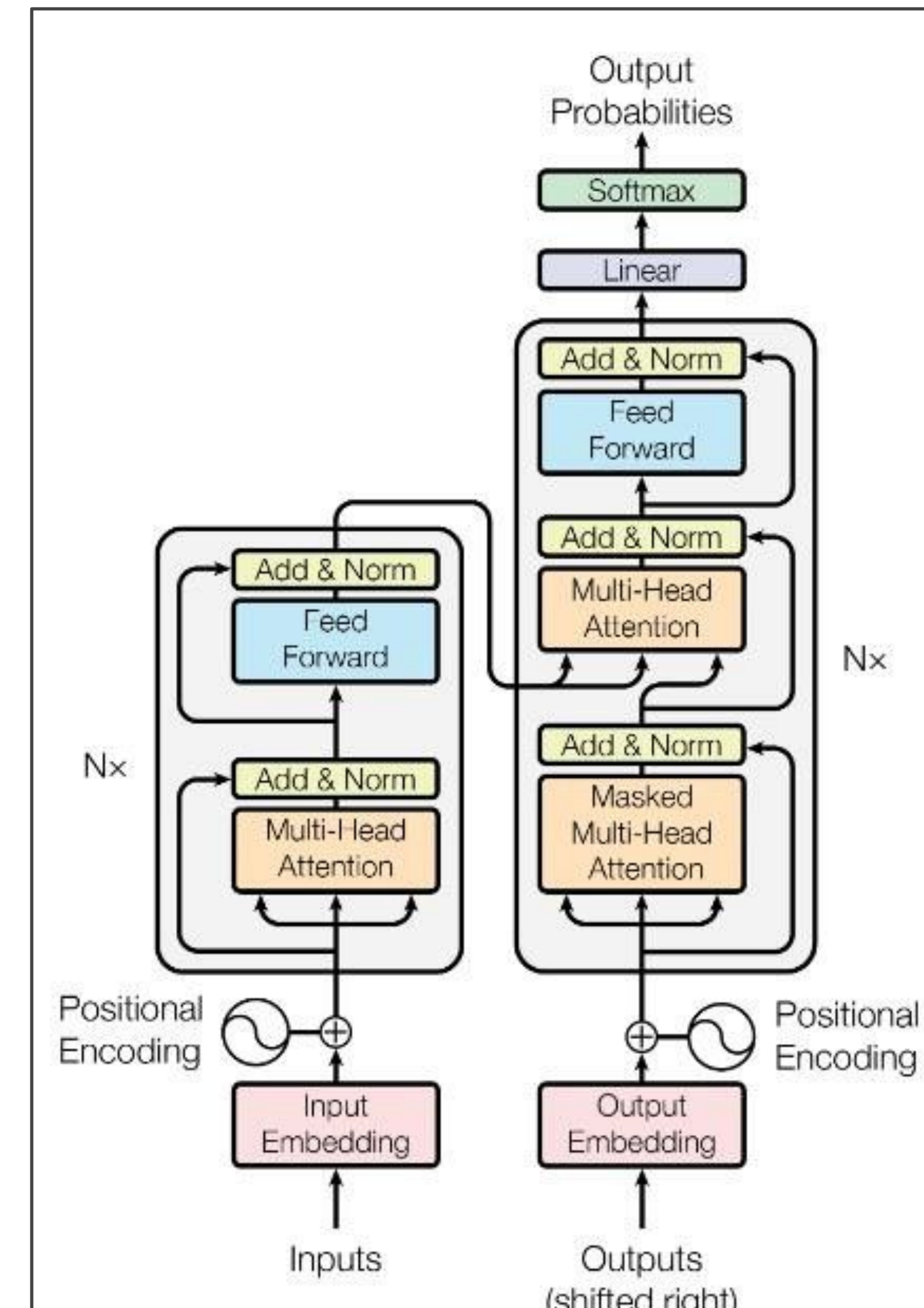


Fig. 1: Facebook's BART Architecture

Conclusion & Future Work

The extension has been published to the google webstore and is currently providing useful privacy policies to various users. It is also cost efficient and is simple to keep running reliably with RunPod. A possible update to make to the extension in the future would be to find contact information for websites without privacy policies anywhere, so that users may receive information directly from the company behind them. Using a more recent NLP model would also allow for more nuanced summaries. Adding support for additional languages is another goal for the future.